

EE 316 - Electronic Design Project

Project: P1

Water Level Controller

Final Project Report

25 June 2026

Objective

To reduce water wastage and eliminate the need for human intervention when pumping water into tanks, it is important to design a simple control system that monitors water levels. This article presents the design and implementation of an automatic water level controller with a digital display. The circuit is low cost, easy to maintain, and operates on a replaceable battery. Our goal is to provide an efficient and effective solution for water level control in tanks using a simple circuit. Whether you are dealing with a domestic water tank or an industrial installation, a water level controller can make your life easier.

Group Members

Common efforts:

Hasan Veyisel Tutar: Capacitor design and manufacturing, real life circuit assembly and testing, and troubleshooting.

Halil Yılmaz: Simulations and circuit building assistance.

Doruk Öğüt: Simulations and circuit building assistance.

Revision History

Week-5 (23.03.2026): We decided what to make the capacitor from.

Week-6 (30.03.2026): We tested our capacitor with the Signal Generator. We progressed as far as the DC signal section.

Week-7 (06.04.2026): We discussed how we could solve our oscillation problem with ICs. We decided on NE555 as the solution.

Week-8 (13.04.2026): We tested our Oscillation Circuit NE555 with electronic lab equipment [1].

Week-9 (20.04.2026): We tried to make progress by combining our oscillation circuit with a capacitor detection section.

Week-10 (27.04.2026): We have decided what we will use as an operational amplifier [6].

Week-11 (04.05.2026): According to our test results, the LM741 integrated circuit did not yield the same results as our simulations.

Week-12 (11.05.2026): Our datasheet research revealed that the LM358 integrated circuit is suitable for the remaining part of our circuit (Amplifier, PWM) [2].

Week-13 (18.05.2026): In the triangular wave section, we obtained a triangular wave by taking the main oscillation circuit, reducing its frequency to a quarter of its original value, and using filtering to account for possible offsets.

Week-14 (25.05.2026): Holiday week, The capacitor was broken, so we redesigned it.

Week-15 (01.06.2026): We completed all our circuits. We conducted troubleshooting for possible offsets.

Week-16 (08.06.2026): We completed all our circuits in the simulation. All the materials have been ordered.

Week-17 (15.06.2026): A demo video was filmed. A band member poured water on the circuit.

Week-18 (22.06.2026): Circuit Completed.

1.Introduction

The aim of this project is to develop an automated water level control system that uses capacitive sensing and feedback systems to reliably and effectively control water levels. The water level control system is designed to determine and manage the water level in a tank using capacitive sensors. When the water level drops below a certain level by the controller, the pump is triggered. When the water level rises above a certain limit, the pump shuts off.

Capacitive sensors, signal processing circuits, and a driver module used to operate a PWM-controlled motor are the system's core components. The dielectric properties of water allow the sensor to detect changes in water level. Resistors and capacitors, including logic gates and analog filters, pass the output from the sensors through various stages.

Another essential part of the system is a feedback loop controlled by an error amplifier. In this loop, the water level and the reference value are continuously compared. The amplifier adjusts control indicators that amplify any discrepancies between the reference value and the water level, thus maintaining the water at the correct level within the system [6].

The output of the error amplifier is used to generate the PWM signal [7]. The capacitive sensor detects the water levels, and the duty cycle of the PWM signal changes the motor operation. When the water level in the tank is below the desired level or completely empty, the motor operates at its maximum duty cycle [10]. As the water level in the tank rises, the duty cycle decreases and the motor switches off. The system also ensures smooth operation of the motor with a current-protected motor driver circuit.

We must overcome three major design challenges by making a water level controller:

Custom Capacitive Sensor Fabrication: Precise level detection necessitates exceptional geometric accuracy. To reduce expenses and prevent measurement inaccuracies associated with the "kettle effect," we utilized SolidWorks and 3D printing to fabricate tailored frames [9]. A continuous, single component aluminum surface was employed to enhance sensitivity, while premium electrical tape acted as a dielectric barrier to inhibit leakage currents and thermal fluctuations.

Signal Conversion through Oscillation: It is necessary to transform the sensor's changing capacitance into a digital signal. Capacitance variations are converted into a frequency modulated square wave using an astable mode NE555 timer [1]. The difficulty is choosing exact $R1$ and $R2$ values to retain high resolution while keeping the output frequency within the microcontroller's

Parasitic Voltage Leakage: A 500 mV leakage was detected at the XOR output during measurement, causing noise and reducing signal reliability. This became one of the project's challenges and required improved grounding, insulation, and filtering.

1.1. Capacitive Water Level Sensors

Our system employs cylindrical capacitive sensors to detect variations in capacitance between two aluminum strips and a plastic tube situated within a tank filled with water. Fluctuations in the water level led the sensors to register changes in capacitance between the electrodes. The higher dielectric constant of water in comparison to air results in an increase in capacitance as the water level rises. We assessed and calculated the capacitance of the capacitor in air and observed how it varied at different water levels. Preliminary experiments indicated that the presence of water on the aluminum surface influenced the conductivity of the capacitor plates. Consequently, we applied a thin layer of plastic wrapping around the capacitive sensors to enhance their sensitivity. This adjustment enabled us to effectively track water levels. We employed the NE555 timer integrated circuit to produce a clock signal with the specified frequency and duty cycle values within the 0 – 5V range.

1.2. Signal Conditioning and DC Conversion

The capacitor's output was converted to direct current (DC) to enable the DC motor to operate effectively [5]. Our clock signal was subsequently used to feed the circuit after it was constructed, functioning as a phase comparison device. The circuit measured the phase difference between our clock signal and the capacitor output, generating a corresponding signal. This was achieved using a comparator comprising two XOR gates; the first digitized our system, while the second compared the capacitor output with the clock signal. The comparator's output was then passed through a low pass filter to produce a DC voltage signal. Prior to utilization, it was necessary to amplify the DC output, as the signal amplitude was initially too low, which risked amplification errors [8].

1.3. Error Amplification

We see that the error amplifier is a circuit that compares the difference between the reference water level and the actual water level. This acts as a feedback loop and keeps the system within its most effective range. When there is a difference between the reference level and the actual level, it continuously monitors and amplifies this difference through the error amplifier [7]. Initially, in the simulation, the supply voltages for the amplifier were set to 5V and GND, and V_{ref} was varied between 0V and 4V. (Note that the maximum amplified DC signal level is 4V. In the experimental setup, a potentiometer is used to adjust the reference voltage. The water level in the bottle is divided into intervals, and the adjustment of the DC signal voltage values is matched to this interval. In this arrangement, the potentiometer sets a reference level voltage, and when the target water level is reached, the error amplifier compares this reference voltage to an actual voltage signal indicating how much water is in the container. Therefore, the motors only pump water when the water level is below the maximum level [6].

1.4 PWM Generation

The square wave generated by the NE555 timer first passes through an *RC* filter circuit to create a delayed version of the original signal. This delayed signal is then converted into a digital signal by using a ground reference XOR gate. After that, another XOR gate compares the original NE555 timer output with the digitized delayed signal. As a result of this comparison, a difference signal is produced at a very small millivolt level. Since this signal is too weak for direct use, it is amplified by a non inverting amplifier stage [6]. After amplification, the signal is applied to a differential amplifier, also called an error amplifier, together with a previously adjusted reference voltage to generate an error signal [7]. This error signal is then sent to the PWM generation stage, where it is compared with a triangular waveform to produce the PWM output. The duty cycle of the PWM signal changes continuously according to the water level, which allows accurate control of both the motor speed and the water flow [10].

1.5 Motor Driver Integration and Protection Circuit

Using PWM to drive the motor via L298 motor driver IC [4]. This integrated circuits design means it can switch effectively and operate safely with the DC motor [5]. The motor speed is directly controlled by the PWM duty cycle, which can be accurately regulated for water pump operation. The system can autorotate motor speed changes and keep water balance by changing the duty cycle according to the water level. This method made the operational safety and functionality of the motor even better.

1.6 Motor and Pump Control

The submersible pump can connect to a motor that runs based on the PWM controlled signal [10]. When the water comes to a maximum, the system reduces duties which slow down or stop pump action. The duty cycle increases when the water level drops below the minimum value, and pump runs to fill up the tank. Also, the present protection circuit is integrated to safeguard likely damage from carrying out such postures through the mode of motor overcurrent circumstances. Move on with the project, at this stage these circuit parts will be implemented over to a breadboard and at each stage signal integrity will be checked and finally check the full system operation under various water level conditions.

2. Technical Description

In order to accurately and continuously monitor the liquid level in the tank, we built this system with a closed-loop control structure. In our design, we sensed the liquid level by placing a capacitive electrode inside the tank. We were able to measure the capacitance change that happens when the liquid level varies by applying a high frequency signal produced by an oscillator from a constant DC source to this electrode. We converted this change obtained in the capacitive sensing block into an electrical signal and then processed it in the amplifier stage to bring it to a suitable level for control. We then compared this signal with a defined reference level in the error amplifier block to obtain the instantaneous error information of the system [7]. By transferring the generated error signal to the PWM generation block, we aimed to create the control signal to be used in the next stage of the system. The next goal of our design is to apply the generated PWM signal to a motor driver circuit to control the pump motor and thus automatically maintain the liquid level in the tank at the desired value. Accordingly, the motor driver, pump, and protection circuit blocks are planned for the later development stages of the system. With this approach, our primary goal is to establish the sensing and control infrastructure, and in the next stage, to transform the system into a fully closed-loop liquid level control structure.

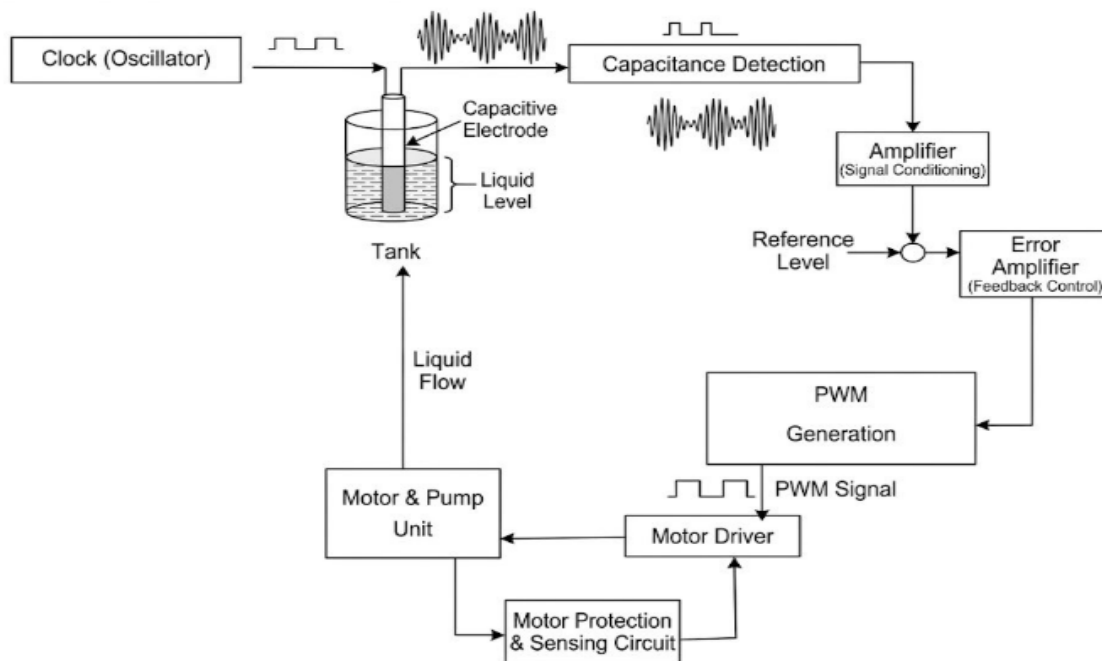


Figure 1. Block Diagram of the Water Level Control System

The system begins by using a clock to create a square wave, which is subsequently transmitted to a capacitor's electrode. In the capacitance detecting part, we also want to produce a DC signal, which requires several intricate procedures.

First, we used a NE555 timer in astable mode to transform the steady DC supply voltage into a steady clock signal [1]. The handcrafted capacitive sensor, whose output changes with capacitance, was then subjected to this clock signal. A phase comparator was used to compare the reference clock and sensor output, and the resulting signal was transformed into PWM for the subsequent motor control stage.

2.1. Capacitor Design

When designing our capacitor, we experimented with a variety of methods. In the end, we decided to create it using a 3D printer. The height and outside diameter of our capacitor were measured theoretically and determined to be 25 cm and 24.8 mm, respectively.

It was found that the cylinder within the capacitor was 28 cm tall. Additionally, the capacitor's inner and outer shell radii were found to be 4.2 mm and 9 cm, respectively, as seen in Figure 2 below.

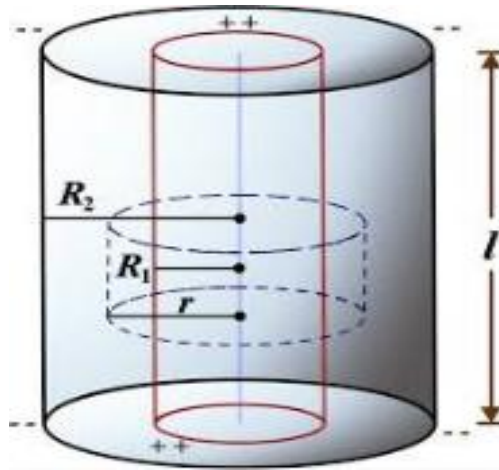


Figure 2. Diagram of cylindrical capacitor

The capacitive sensor was built using aluminum tape on the external surfaces of the inner and outer cylindrical shells that formed the electrode structure of the sensor. This arrangement provides a measurable capacitance between the two conductive layers that varies according to the surrounding dielectric medium and enables liquid level detection as shown in Figure 3 below.



Figure 3. Picture of the Capacitor

According to our research, the formula for the capacitance of a cylindrical capacitor is given

$$C = \frac{2\pi\epsilon_0 * \epsilon_m * L}{\ln(b/a)} \quad (1)$$

C is the capacitance of the cylinder, a is the inner radius of the cylinder, L is the length of the cylinder, b is the outer radius of the cylinder, ϵ_0 is the permittivity of free space, and ϵ_m is the permittivity of medium.

$$C_{water} = \frac{2\pi\epsilon_0 * \epsilon_{water} * h}{\ln(b/a)} \quad (2)$$

$$C_{air} = \frac{2\pi\epsilon_0 * \epsilon_{air} * (L-h)}{\ln(b/a)} \quad (3)$$

Then the total capacitance is:

$$C_{total} = C_{water} + C_{air} \quad (4)$$

$$C_{total} = \frac{2\pi\epsilon_0 * \epsilon_m}{\ln(b/a) * (\epsilon_{water} * h + \epsilon_{air} * (L-h))} \quad (5)$$

2.2 Oscillation Generation

Based on our preliminary research and design requirements, we selected the TLC555 timer to generate the excitation signal for the sensing stage. Due to its simple implementation, stable operation, and ability to produce a continuous square wave without external triggering, the TLC555 was chosen as the oscillator source [1]. Configured in astable mode, it generates a stable periodic signal that is used as the reference excitation input for the capacitive sensing circuit .

TLC 555 Astable Multivibrator

The TLC555 astable multivibrator circuit shown in Figure 4 generates a clock signal, V_{CLK} , which is a 35 kHz square wave oscillating between 0 V and 5 V, with approximately a 50% duty cycle. This clock signal provides a stable timing reference essential for DC signal generation in our design [1].

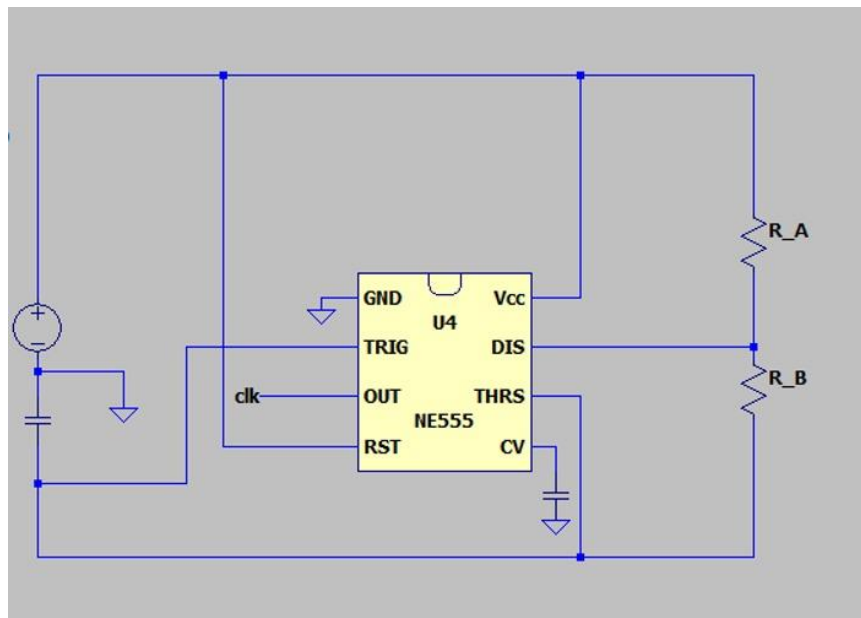


Figure 4. TLC555 astable oscillator generates the reference square-wave signal

The Frequency is calculated with this formula:

$$frequency = 1.44 * (2 * R_A + R_B) / C_{timing} \quad (6)$$

And the duty cycle is calculated with:

$$Duty\ Cycle = \frac{(R_A + R_B)}{(2 * R_A + R_B)} \quad (7)$$

2.3 Capacitance Detection

At this point, we employed a straightforward RC network to monitor the charge discharge behavior of our capacitive sensor using the produced square wave signal. Based on the capacitor's charging and discharging properties, a phase-shifted waveform was produced when it reacted to the excitation signal. In order to stabilize the waveform and retrieve its phase-dependent behavior, this signal was then processed utilizing XOR-based phase comparison stages.

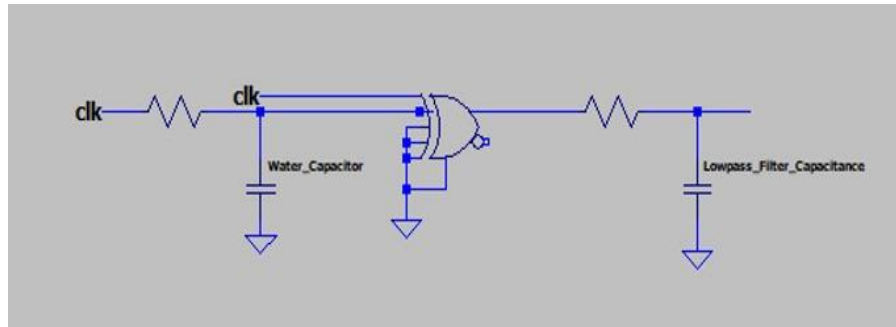


Figure 5. Circuit Diagram of Capacitance Detection

Delayed Signal Generation

Based on our preliminary research, we found that the simplest way to observe the capacitance variation of our handmade sensor was to examine its charge discharge behavior in a basic RC structure. By applying the generated square wave signal to the sensor, we obtained a delayed output due to the capacitor's charging and discharging response. This phase shifted waveform allowed us to observe the capacitance effect directly in the time domain

The delay is determined by the RC time constant,

$$T = RC$$

and the delayed output is described by

$$V_{out}(t) + RC \frac{dV_{out}}{dx} = V_{in}(t)$$

Signal Digitalization and Convert DC

Based on our analysis, the delayed analog signal needed to be converted into a more stable and measurable form for evaluation. To achieve this, we compared the delayed capacitor response with the reference clock signal using XOR-based phase comparison. Since the XOR gate (74LS86) produces HIGH output only when its inputs differ, the phase difference between the delayed signal and the reference signal was translated into a pulse-width-varying digital waveform [2].

The digital output was processed to obtain a stable, DC equivalent signal proportional to the detected phase shift. By converting the phase-dependent waveform into a DC level, a more stable representation of the capacitance variation was achieved. This enabled the sensor response to be observed as a measurable voltage level, thereby simplifying the evaluation of capacitance dependent behavior.

Filtering

After the XOR stage, the output signal contains high-frequency switching components and must be filtered to obtain a stable DC level. For this purpose, an RC low-pass filter is used to suppress the high-frequency ripple while preserving the average value of the signal. This allows the phase-dependent digital waveform to be converted into a smoother and more stable DC output.

The cutoff frequency of the filter is determined by:

$$f_c = \frac{1}{2\pi RC}$$

This relation was used to select the filter parameters and evaluate the response of the filtering stage.

Capacitance Detector Design Targets			
Description of design target	Min.	Max.	Unit
Delayed signal output voltage	0	4.96	V
DC signal output voltage	0	578	mV
Filtering Frequency	10	23	Hz
Capacitance Value	1.5	5.15	nF

Table 1. Capacitance Detection Design

2.4 Amplifier

Following the theory learned from the principles of analog amplifiers in Circuit Analysis I, an operational amplifier was employed to boost the voltage output after filtration.

A high pass filter was designed to limit any noise in low frequencies prior to amplifying the input signal.

Because the output voltage from the capacitance detection stage was too low for comparison purposes, a non-inverting amplifier was chosen to boost the output signal voltage while keeping its polarity intact.

The output voltage is expressed by the equation below:

$$V_{out} = V_{in} \left(1 + \frac{R_f}{R_g} \right) \quad (10)$$

Using the resistor values chosen above, the gain was controlled to achieve the desired output voltage. The circuit design is depicted in Figure 6. The output voltages are provided in the table below.

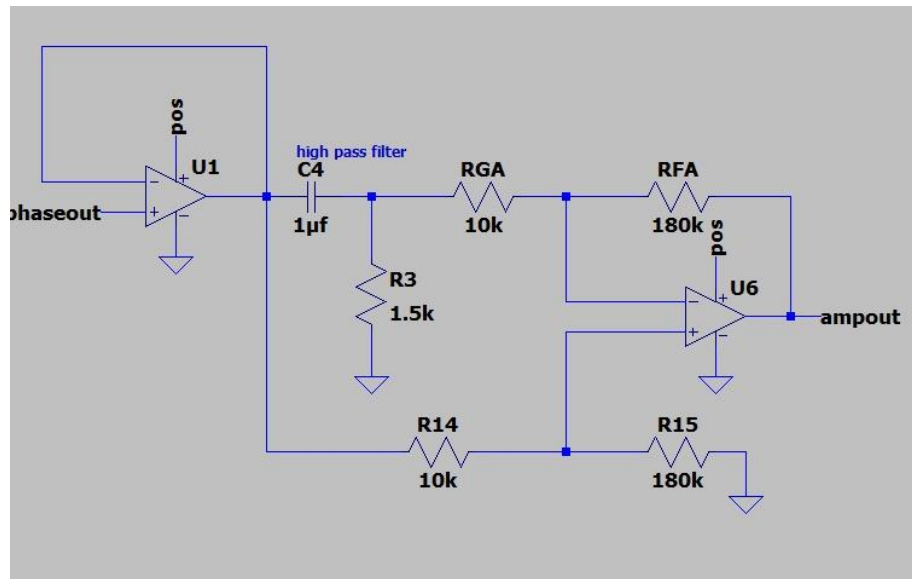


Figure 6. Circuit diagram of Amplifier

Amplifier Design Targets			
Description of design target	Min.	Max.	Unit.
DC Voltage	170	700	mV
Output voltage for non-inverting amplifier	3	4.2	V
Gain	7.2	56.44	-

Table 2. Amplifier design targets and parameters

2.5 Error Generation and Level Adjustment

Here, the error amplifier is employed to compare the amplified sensor output with a variable reference voltage. This reference voltage is connected to the non-inverting terminal of the operational amplifier via the voltage divider formed using a potentiometer, whereas the amplified DC signal is fed to the inverting terminal via resistor R5. The potentiometer can be used to set the required level of water manually. Thus, the output of the error amplifier circuit is the difference between the sensed and reference levels of water.

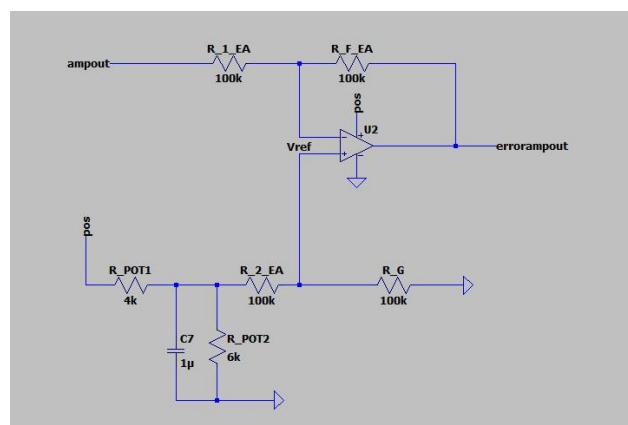


Figure 7. Error Amplifier

Error Amplifier Design Targets			
Description of design target	Min.	Max.	Unit
DC voltage (from the amplifier)	1	5.1	V
Output Voltage	0.234	4	V
Reference Voltage	1.03	4.5	V

Table 3. Design Targets for the Error Amplifier

The equation for output voltage from the error amplifier will be:

$$V_{out} = \left(1 + \frac{R_f}{R_1}\right)V_+ - \left(\frac{R_F}{R_1}\right)V_{in} \quad (11)$$

Where in the circuit $\frac{R_F}{R_1} = \frac{100k\Omega}{100k\Omega} = 1$ So the equation will become,

$$V_+ = \frac{1}{2} V_{ref} \quad (12)$$

Finally, the output equation can be simplified as:

$$V_{out} = V_{ref} - V_{in} \quad (13)$$

It means the error amplifier provides its output according to the difference between selected water level reference and the sensor voltage

2.6 Triangular Wave and PWM Generator

Two cascaded 74LS74 D-type flip-flops and an RC wave-shaping network were used in the revised design to create the triangular reference waveform. After two stages, the output frequency is as follows because each flip-flop divides the input clock frequency by two:

$$f_{out} = \frac{f_{clk}}{4} \quad (14)$$

After that, the divided square-wave signal was fed into a circuit with parallel high-pass and low-pass filter branches and a DC route. The square wave was transformed into a roughly triangular waveform by the combined response of these branches. Each RC filter's cutoff frequency is determined by

$$f_c = \frac{1}{(2\pi RC)} \quad (15)$$

The error amplifier output and the triangular reference waveform were compared during the PWM generation stage. When v_{error} exceeds $v_{triangle}$, the PWM output

is HIGH; when v_{error} falls below $v_{triangle}$, it is LOW. As a result, while the frequency of the PWM signal stays constant, changes in the error amplifier output alter its duty cycle.

$$Duty\ Cycle\ (\%) = \frac{Pulse\ Width}{(Period \times 100)} \quad (16)$$

The figure and tables below show the triangular-wave generation circuit, PWM output, and related measurement results.

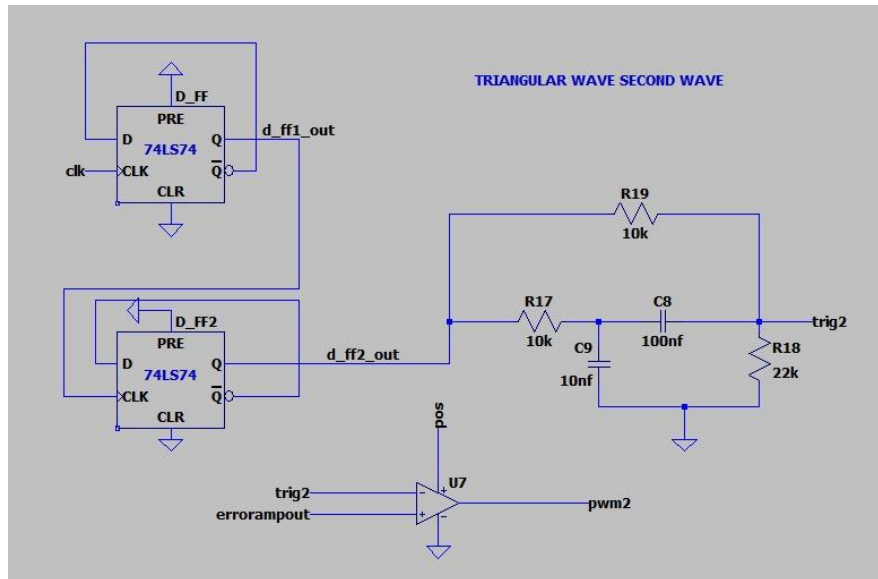


Figure 8. Triangle Signal Generator and PWM

Clock Signal for triangular wave Design Targets			
Description of design target	Min.	Max.	Unit
Input voltage	4.95	5.05	V
Output low voltage	0	28.26	mV
Output high voltage	4.95	4.99	V
Output duty cycle	84	88	%
Output frequency	9	9.98	kHz

Table 4.Design targets for the PWM triangular wave

2.7 Motor Drive

In this case, the L298 DC and stepper motor driver is employed to operate the motor. The L298 is a dual H-Bridge motor driver that is widely used for driving DC and stepper motors. In this setup, the L298 driver is used to regulate the speed of the DC motor using the output PWM signals.

Generally, the L298N driver can support relatively high current and voltages of up to 2A per channel and 46V. Besides, the L298N comes with protection circuits such as over-current and thermal shutdown which makes the circuit operation safe. The chosen DC motor runs at either 12V/3A or 6.5V/2A. The voltages and currents are within the range of the chosen motor driver module [4].

Schematic Diagram:

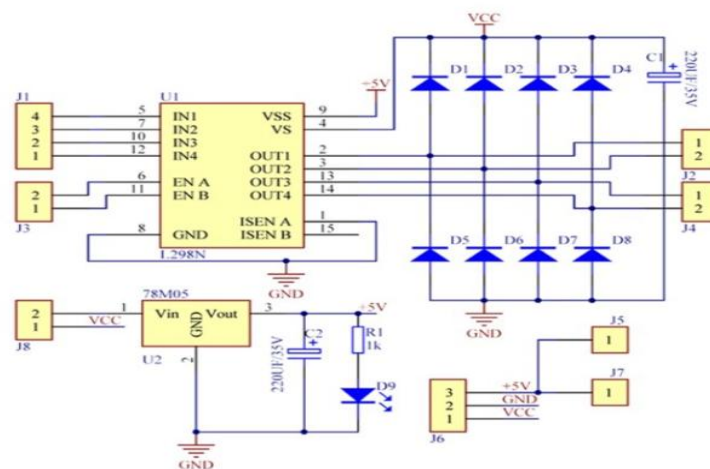


Figure 9. The pinout configuration of L298N Module

The procedure starts with generating a square wave clock signal. This signal is provided to the capacitive electrode, and the capacitance value is altered in response to the level of water in the tank. Besides, the whole circuit should generate a DC signal after passing through several processing operations in the capacitive detection circuit.

At first, the clock signal is generated by the NE555 timer in astable mode. The clock signal then goes through the handmade capacitive sensor, and the output from it depends on the capacitance changes. Next, the output of the clock signal and the output of the sensor are connected to the phase comparator. The output from it becomes the PWM signal, thus allowing varying the duty cycle.

2.8 DC Motor

A 12 V DC brushless submersible motor is chosen in this work to pump the water and to control the level of water in the tank. DC motors run on DC power and can be energized by a DC voltage source. The chosen motor was tested in the lab and runs properly with 12 V/3 A or 6.5 V/2 A.

The structure of a basic 12 V DC motor is illustrated in figure 9. The basic elements of the motor are rotor with permanent magnets and stator with electromagnetic windings.

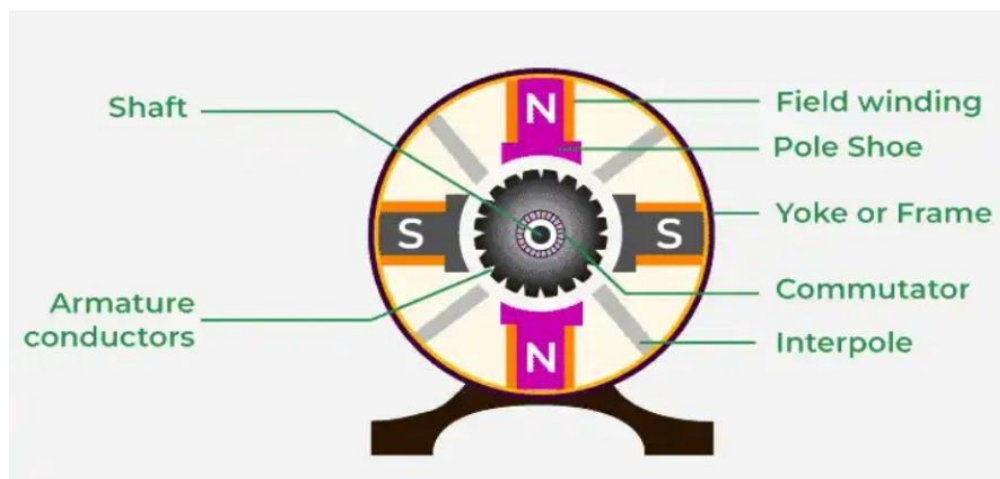


Figure 10. DC motor mechanics scheme

DC motors are perfect for tasks requiring adjustable speed and torque characteristics. In this project, the motor speed is controlled via the change in PWM duty cycle. Thus, the water flow can be regulated depending on the level of water in the tank [6]

2.9 Motor Protection Circuit

The ACS712 is a Hall effect current sensor that generates output voltage directly proportional to the current passed through it. In this case, the ACS712 current sensor measures the current consumed by the DC motor. The output voltage of the current sensor is supplied to the inverting input terminal of the LM393P comparator whereas the non-inverting input terminal receives the reference voltage set as the ACS712 output voltage for the current of 2 A.

The comparator compares the motor current with the threshold level. The output of the comparator will be high when the measured current is less than 2 A. The output of the comparator is used to drive the MOSFET gate voltage close to 5V in this case. The motor operation will be enabled by turning on the MOSFET. However, when the current is greater than 2 A, the output of the comparator will go low disabling the enable signal of the motor driver circuit.

Moreover, a Schmitt Trigger stage is used in order to achieve cleaner switching performance. Without it, there may be an unstable output or output that remains in a state somewhere in between HIGH and LOW state of the comparator output depending on how much the input voltages differ. These uncertain states of output may not be enough for a full activation of MOSFET. This situation is avoided by using the Schmitt Trigger since its output will definitely be in either HIGH or LOW state only.

3 Test Results

3.1 Oscillation Generation

The measurements corresponding to the oscillation signal generated by 555 timer circuit are given in the figure and following table.

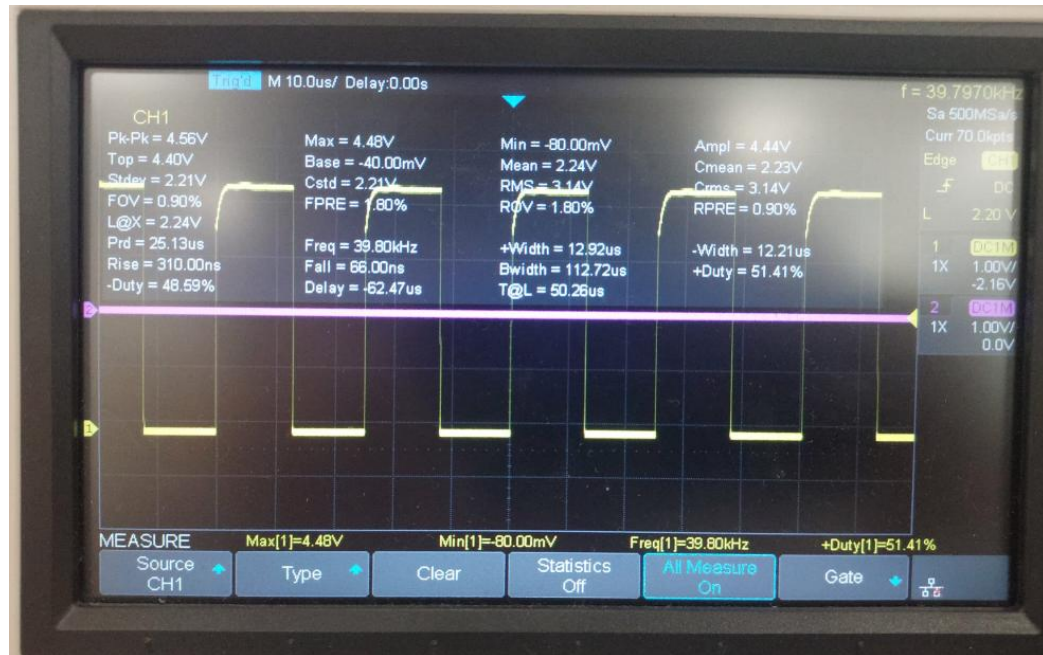


Figure 11. The output of NE555 Timer Circuit

Oscillation Signal Design Results		
Design Targets	Results	Unit
Input voltage	5.1	V
Output low voltage	-80	mV
Output high voltage	4.48	V
Output duty cycle	48.59	%
Output frequency	39.80	kHz

Table 5. Oscillation Circuit Result

The simulation yielded a value close to what was tested. However, the actual result is always different.

3.2 Capacitance Detection

The minimum capacitor values and low-pass filter results obtained are shown in the table and photographs.

Capacitance Detector Design Results			
Design Targets	Min.	Max.	Unit
Delayed signal output voltage	-60	3.6	V
DC signal output voltage(LPF)	60	250	mV
Filtering Frequency (Capacitor)	35.4	38.9	kHz
Capacitance Value	1.1	5.4	nF
Filtering Frequency (for DC signal)	35	46	Hz

Table 6. Capacitance Detection Part Results

Now, the low-pass outputs are 10%, 50%, and 90%, respectively.

- For %10



- For %50



- For %90



3.3 DC Offset Amplifier and Level-Adjustment

Our dc amplifier and error generation results are shown below. First amplifier table,

DC Offset Amplifier Design Results			
Design targets	Min.	Max.	Unit
DC voltage	60	250	mV
Output voltage for non-inverting amplifier	0.987	4.347	V
DC Signal gain	18	30	-

Table 7. Amplifier Circuit Results

We determined that the LM741 was not suitable for our circuit. We had initially considered using the LM741 when designing the circuit. However, our tests revealed that LM358 was a more suitable alternative [2].

Our error amplifier figure and table are now below.



Figure 12. Error signal is yellow

Error Amplifier Design Results			
Design targets	Min.	Max.	Unit
DC voltage (from the non-inverting amplifier)	0.905	4.352	V
Reference voltage	1.02	4.6	V
Output voltage	0.045	3.57	V
Reference Resistor	1.05	9.10	k Ω

Table 8. Error Amplifier Results

3.4 Triangular Wave and PWM Generation

We reduced the frequency of our signal from the main oscillator using a 74ls74 in real-world scenarios. The test results are shown in the figure and table below.



Figure 13 Triangular wave and PWM Signals

Triangular wave and PWM Design Results		
Design targets	Measured	Unit
Triangular low voltage	104.5	mV
Triangular high voltage	3.53	V
Triangular duty cycle	92.6	%
Triangular wave frequency	9.08	kHz
Output high voltage (PWM)	4.56	V

Table 9. Triangular wave and PWM Results

When the error signal voltage is higher than triangular voltage, the PWM signal becomes HIGH. In real life, some noise may have occurred. It had no effect; better conditions can be created by moving the circuit in the air or from side to side.

4. Measurement and Simulation Methods

Figure and Description	Measured Quantity	Equipment / Tool	Explanation / Method
Figure-3: Picture of capacitor	Capacitance(nf)	Multimeter	We measured it with a multimeter.
Figure -14: TLC555 astable oscillator generates the reference square-wave signal	Clock Frequency(khz), Duty Cycle(%)	Ltspice	A circuit was created and measurements were taken using Ltspice.
Figure- 15: Triangle Signal Generator and PWM	Triangular Peak Voltage (V)	Ltspice	It was tested and found to give 5V at points where the error voltage was greater than the triangle wave.
Figure-16:The output of NE555 Timer Circuit	Frequency (kHz)	Digital Oscilloscope	The oscillation frequency measured as 39.80 kHz.
Capacitance Detection Figures	Switching Voltage mV	Digital Oscilloscope	60mV was measured at 10% and 250mV at 90%.
Figure 17 Triangular wave and PWM Signals	PWM voltage (V)	Digital Oscilloscope	At points where the error signal was high, the PWM voltage was measured at 4.56 V.

5. Conclusion

Automatic water level control system was designed and implemented by our team. In this design, the variations in water level are converted to capacitance variations. The capacitance variations are further processed using phase comparison and filtering techniques to obtain a DC signal. The signal obtained is amplified using an error amplifier and the output is compared with triangular wave form to obtain a PWM signal. The PWM signal controls the speed of submersible DC motor using motor driver circuit.

Improved design of capacitive sensor has been done for better measurements and sensitivity. The capacitive sensors having cylindrical shapes made with aluminum tape electrodes have been employed and the electrodes have been insulated with insulation layer to avoid direct contact of water.

The feedback control scheme of the system is mostly done with the help of the error amplifier output. The error voltage changes depending on the difference between the actual water level and the reference level. This helps adjust the PWM duty cycle which in turn helps change the speed of the motor depending on the water level and keep the water level at the required value. Moreover, an overcurrent protection circuit was added with the help of a current sensor and comparator.

During the course of the project, both simulations and practical circuit tests were conducted. Some problems arose during the project such as problems with the conductivity of the capacitive sensor and the calibration of the error amplifier circuitry. The problem with the conductivity of the capacitive sensor was solved through the insulation of the capacitor.

In summary, through this project, our group gained firsthand experience in analog electronics, feedback systems, PWM creation, sensor development, and overcurrent protection circuitry. The resulting system includes numerous vital components, such as the homemade cylindrical capacitor, clock signal generator, RC filters, error amplifier, PWM generator, motor controller, and overcurrent protection circuit. All these components together comprise an automatic water level control system [10].

6. Component List and Cost

Component description	Part Number	Manufacturer	Supplier	Cost (TRY)
Timer/Clock	TLC 555	Texas Instruments	cakirelektronik.com	10.6
Operational amplifier	LM358P	Texas Instruments	cakirelektronik.com	7.96
Logic gate (XOR gate)	74LS86	STM Philips	cakirelektronik.com	11.89
DC Motor Driver Module	L298N	GlintDeer	cakirelektronik.com	104.2
D Flip Flop	74LS74	Texas Instruments	cakirelektronik.com	12.5

The total cost of the necessary hardware parts needed in this project is estimated at 200 TRY. As this system involves the use of mostly inexpensive electronic parts like op-amp, logic gate, timer IC, and motor driver module, therefore it is economical. Almost all of the parts can be purchased locally from the websites of electronics suppliers like Direnc.net and Robotistan. The costliest part in the given list is the L298 motor driver module which is inexpensive enough to drive the DC motor in this project. Hence, the water level control system designed is low-cost and feasible [2].

7. Project Management

Project Steps	Project Member#1 (Hasan Veysel TUTAR)	Project Member#2 (Halil Yılmaz)	Project Member#3 (Doruk Öğüt)
555 Timer design and simulation	X	X	X
555Timer experiment setup and test	X	X	
Capacitor Design	X	X	
Capacitor Setup and test	X		
PWM design and Simulations		X	X
PWM Setup and Test	X		
Low pass design and simulations		X	X
Low Pass setup and Test	X		
Error Amplifier Design and Simulations		X	X
Error Amplifier Setup and Test	X		X
Triangular Wave Design and Simulations		X	X
Triangular Wave Setup and Test	X		
DC Amplifier design and Simulations	X	X	X
DC Amplifier Setup and Test	X		
Lab Attendance	XX	X	X
Parts Procurement	X	X	X
Report Writing	X	X	X
Motor Driver Setup and Test	X		X
Laboratory Support	X	X	XX

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